

Q: How to define a distance
between two polynomials. (function)

1. geometry $\rightsquigarrow$ topological structure

2. number $\rightsquigarrow$ Algebraic structure

3. order $\longrightarrow$ ordered structure

$$d(f, g) = \sqrt{\sum_{i=0}^m (a_i - b_i)^2}$$
$$\text{L } f = \sum_{i=0}^m a_i x^i \quad g = \sum_{i=0}^m b_i x^i$$

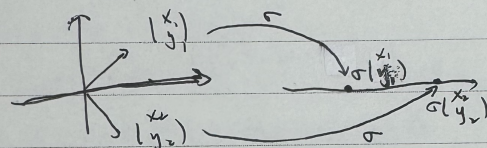
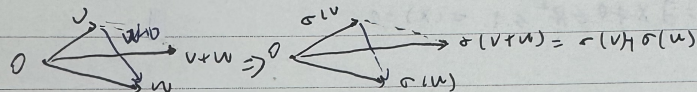


§2. Linear Maps (Linear Transformations) $V = \{\text{vector space}\} / \mathbb{F}$

Given $U, W \in V$, Define a map $\sigma: V \rightarrow W$ satisfying:

- 1) $\sigma(U+V) = \sigma(U) + \sigma(V)$ $\sigma(xU+x'V) = x\sigma(U) + x'\sigma(V)$
- 2) $\sigma(\lambda U) = \lambda\sigma(U)$ $\sigma: A \text{ linear map from } V \text{ to } W$

• Try to find the geometric meaning



linear map = linear function

$$f\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right) = f\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}\right) + f\left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}\right)$$

$$\mathbb{R}^n \xrightarrow{f} \mathbb{R}^m \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$$

$$\mathbb{R}^2 \xrightarrow{f} \mathbb{R}^3 \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^2, \mathbb{R}^3)$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto f\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = f\left(\begin{pmatrix} x \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ y \end{pmatrix}\right) = f\left(\begin{pmatrix} x \\ 0 \end{pmatrix}\right) + f\left(\begin{pmatrix} 0 \\ y \end{pmatrix}\right) = x f\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) + y f\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = ax + by = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$v \mapsto f(v) = (f(e_1) \ f(e_2)) v = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}_{2 \times 2} v = A_{2 \times 2} v$$

Conclusion $\forall f \in \text{Hom}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R}^m)$ can be written as $f(v) = A_{m \times n} v = 1 \cdot v$

Thm. $\text{Wich} = \dim V = n, \dim W = m \quad \forall \sigma \in \text{Hom}_{\mathbb{R}}(V, W)$ can be

written as $\sigma(v) = (w_1, w_2, \dots, w_m)(AX)$

$$v = (v_1, v_2, \dots, v_n) \quad X \quad x \in \mathbb{F}^n$$

$$\sigma(v) = \sigma\left(\sum x_i v_i\right) = \sum x_i \sigma(v_i) = (\sigma(v_1), \sigma(v_2), \dots, \sigma(v_n)) X$$

$$\sigma(v_j) = a_{1j} w_1 + a_{2j} w_2 + \dots + a_{mj} w_m = (w_1, w_2, \dots, w_m) \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

$$\sigma(v) = [(w_1, w_2, \dots, w_m) A_{m \times n}] X = (w_1, w_2, \dots, w_m) [A_{m \times n} X]$$

compare sets:

1. $f: A \rightarrow B$ surjection = 满射 $\forall b \in B, \exists a \in A \text{ s.t. } f(a) = b \Leftrightarrow |A| \geq |B|$

2. $g: A \rightarrow B$ injection = 单射 $a_1 \neq a_2 \in A \Rightarrow f(a_1) \neq f(a_2) \Leftrightarrow |A| \leq |B|$
 $[f(a_1) = f(a_2) \Leftrightarrow a_1 = a_2]$

3. $f: A \rightarrow B$ ^{双射} bijection: surjection + injection

Def. Let $V, W \in \mathcal{V}$, Then V and W are isomorphic (同构) if

$\exists \sigma \in \text{Hom}(V, W)$ s.t. σ is an isomorphism i.e. σ is a bijection

eg. $\mathbb{R}^2$ and $\mathbb{R}$ are isomorphic as sets

but $\forall \sigma \in \text{Hom}(\mathbb{R}^2, \mathbb{R})$ is not an isomorphism (as v.s.)

$\hookrightarrow$ injection: $\exists x \neq 0 \in \mathbb{R}^2$ s.t. $\sigma(x) = 0$

